

Name

In[2]:= **Expand**[$2 (r + 2)^2 - 25$]

Out[2]= $-42 + 8 r + 2 r^2$

- For the ODE $2 y'' + 8 y' - 42 y = e^{3t}$

- Write down the complementary sol.

$y_c(t) =$

- Compute a particular sol.

$y_p(t) =$

- The Aug Mat for $y(0) = 2, y'(0) = -3$ is

1. For the ODE $2 y'' - 4 y' - 96 y = t + 1$

- 1.1. Write down the complementary sol.

$y_c(t) =$

- 1.2. Compute a particular sol.

$y_p(t) =$

- 1.3. The Aug Mat for $y(0) = 4, y'(0) = -2$ is

- For the ODE $2y''' - 24y' + 32y = e^{2t}$

- Write down the complementary sol.

$$y_c(t) =$$

- Compute a particular sol.

$$y_p(t) =$$

- The Aug Mat for $y(0) = -2, y'(0) = 1, y'''(0) = 1$ is

$$A =$$

2. For the ODE $2y''' + 2y'' - 2y' + 30y = e^{-t}$

- 2.1. Write down the complementary sol.

$$y_c(t) =$$

- 2.2. Compute a particular sol.

$$y_p(t) =$$

- 2.3. The Aug Mat for $y(0) = 1, y'(0) = 2, y'''(0) = 4$ is

$$A =$$